

EP2108 - Special Theory of Relativity

Assignment 2

Instructions: Answer all questions. Numerical answers should include appropriate units. Use $\gamma = (1 - v^2/c^2)^{-1/2}$ throughout.

1 Lorentz Transformation of Coordinates

- 1.1** A train of proper length 1000 m moves at $v = 0.5c$ relative to the ground. Lightning strikes both ends of the train simultaneously as measured in the ground frame.
- (a) Find the time interval between the two strikes as measured in the train's frame.
 - (b) Determine which end of the train (front or back, in the direction of motion) is struck first according to an observer on the train.
 - (c) Verify that the spacetime interval between the two strike events is the same in both frames.

2 Lorentz Transformation of Velocity

- 2.1** Two spacecraft travel in the *same* direction relative to Earth: spacecraft A at $0.90c$ and spacecraft B at $0.60c$, with B trailing behind A . Find the speed of A as measured by an observer on B .
- 2.2** Two particles move directly toward each other, each with speed $0.9c$ as measured in the lab frame. Find their relative speed of approach as measured by an observer riding on one of the particles.

3 Length Contraction

- 4.1** A spacecraft has a proper length of 120 m. As measured by an observer on Earth, the spacecraft is 72 m long. Find its speed relative to Earth.
- 4.2** A meter stick at rest in frame S' makes an angle of 30° with the x' -axis. Frame S' moves at $v = 0.6c$ relative to S along the common x -axis. Find the length of the stick and the angle it makes with the x -axis as measured in frame S .

4 Time Dilation

- 5.1** A cosmic-ray particle travels between two detectors separated by 10 km, as measured in the lab frame, in a time of $40 \mu\text{s}$ (also measured in the lab frame). Find (a) the speed of the particle relative to the lab, and (b) the proper time experienced by the particle between the two detectors.
- 5.2** GPS satellites orbit at a speed of about 3.87 km/s relative to the Earth's surface. Estimate the special-relativistic time dilation (in μs per day) of the satellite clocks relative to clocks on the ground.

5 Relativistic Mass

- 6.1** Explain the concept of relativistic mass $m = \gamma m_0$, where m_0 is the rest mass, and discuss briefly why modern treatments generally prefer to keep mass invariant instead.
- 6.2** An electron ($m_0 = 9.11 \times 10^{-31} \text{ kg}$) is accelerated to $v = 0.99c$. Compute its relativistic mass and compare it to its rest mass. At what speed does the relativistic mass equal $10 m_0$?

6 Relativistic Energy and Momentum

- 7.1** Starting from the relativistic momentum $\mathbf{p} = \gamma m_0 \mathbf{u}$, derive the energy-momentum relation

$$E^2 = (pc)^2 + (m_0 c^2)^2.$$

- 7.2** A proton ($m_0 c^2 = 938 \text{ MeV}$) is accelerated to a kinetic energy of 2000 MeV. Find (a) its total energy, (b) its momentum (in MeV/c), and (c) its speed as a fraction of c .
- 7.3** A particle of rest mass m_0 decays at rest into two identical photons. Using conservation of energy and momentum, show that each photon carries energy $m_0 c^2/2$ and that they are emitted in exactly opposite directions.

7 Relativistic Doppler Effect

- 8.1** Derive the relativistic Doppler shift formula for a source moving directly away from an observer, with recession speed v :

$$f_{\text{obs}} = f_{\text{source}} \sqrt{\frac{1 - v/c}{1 + v/c}}.$$

- 8.2** A distant galaxy emits light at a rest-frame wavelength of 656.3 nm (H_α line) and is observed on Earth at 685.0 nm. Find the galaxy's speed relative to Earth.

- 8.3** A spaceship approaches a space station at $v = 0.6c$ and emits a radio signal of frequency 10 GHz. Find the frequency detected by the station (i) while approaching, and (ii) immediately after passing and receding, at the same speed.